

Assignment 3

2026-06-11

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(caret)
```

```
## Loading required package: ggplot2

## Loading required package: lattice
```

```
library(e1071)
```

```
##
## Attaching package: 'e1071'

## The following object is masked from 'package:ggplot2':
##
##   element
```

```
UniversalBank<- read.csv("UniversalBank.csv")

my_data<- UniversalBank
```

```
colnames(my_data)
```

```
## [1] "ID"           "Age"           "Experience"
## [4] "Income"       "ZIP.Code"      "Family"
## [7] "CCAvg"        "Education"     "Mortgage"
## [10] "Personal.Loan" "Securities.Account" "CD.Account"
## [13] "Online"       "CreditCard"
```

Part A

```
set.seed(123)
```

```
train_index<- createDataPartition(my_data$Personal.Loan, p = 0.6, list = FALSE)
```

```
training_data<- my_data[train_index,]
```

```
validation_data<- my_data[-train_index,]
```

```
pivot_table<- ftable(CreditCard = training_data$CreditCard, Personal.Loan = training_data$Personal.Loan
```

```
pivot_table
```

```
##               Online    0    1
## CreditCard Personal.Loan
## 0           0           785 1145
##           1           65  122
## 1           0          317  475
##           1           34   57
```

Row variables: CreditCard and Personal.Loan Column variables: Online Cell values represent the number of customers in each category combination.

Part B

```
57/(475 + 57)
```

```
## [1] 0.1071429
```

$P(\text{Personal.Loan} = 1 \mid \text{Online} = 1, \text{CreditCard} = 1)$ Numerator: 57 customers had a credit card, used online banking, and accepted a personal loan. Denominator: 532 customers had a credit card and used online banking. The probability that a customer who owns a bank credit card and is actively using online banking services will accept the loan offer is $57/532 = 0.1071$, or 10.71%

Part C

```
online_pivot<- table(Loan = training_data$Personal.Loan, Online = training_data$Online)
```

```
online_pivot
```

```
##      Online
## Loan    0    1
##    0 1102 1620
##    1   99  179
```

```
creditcard_pivot<- table(Loan = training_data$Personal.Loan, CreditCard = training_data$CreditCard)
```

```
creditcard_pivot
```

```
##      CreditCard
## Loan    0    1
##    0 1930  792
##    1  187   91
```

Part D

```
91 / (187 + 91)
```

```
## [1] 0.3273381
```

```
179 / (99+179)
```

```
## [1] 0.6438849
```

```
278 / 3000
```

```
## [1] 0.09266667
```

```
792 / (1930 + 792)
```

```
## [1] 0.2909625
```

```
1620 / (1102 + 1620)
```

```
## [1] 0.5951506
```

```
2722 / 3000
```

```
## [1] 0.9073333
```

$P(CC = 1 \mid Loan = 1)$ is 0.3273 or 32.73% $P(Online = 1 \mid Loan = 1)$ is 0.6439 or 64.39% $P(Loan = 1)$ is 0.0927 or 9.27% $P(CC = 1 \mid Loan = 0)$ is 0.291 or 29.10% $P(Online = 1 \mid Loan = 0)$ is 0.5952 or 59.52% $P(Loan = 0)$ is 0.9073 or 90.73%

Part E

```
loan1<- (91 / (187 + 91)) * (179 / (99 + 179)) * (278 / 3000)
```

```
loan0<- (792/(1930 + 792)) * (1620 / (1102 + 1620)) * (2722 / 3000)
```

```
loan1 / (loan1 + loan0)
```

```
## [1] 0.1105637
```

$P(Loan = 1 \mid CC = 1, Online = 1)$ is 0.1106 or 11.06%

Part F

The Naive Bayes probability gave 11.06% while the probability obtained from the pivot table in Part B was 10.71%. Since the Naive Bayes estimate assumes independence between variables, the pivot table value is generally the more accurate estimate. It comes directly from the data without the independence assumption.

Part G

The entries needed for computing $P(\text{Loan} = 1 \mid \text{CC} = 1, \text{Online} = 1)$ are: $P(\text{CC} = 1 \mid \text{Loan} = 1)$ $P(\text{Online} = 1 \mid \text{Loan} = 1)$ $P(\text{Loan} = 1)$ $P(\text{CC} = 1 \mid \text{Loan} = 0)$ $P(\text{Online} = 1 \mid \text{Loan} = 0)$ $P(\text{Loan} = 0)$

These were obtained from the pivot tables created in Part C and were used in the Naive Bayes calculation in Part E.

```
nb_model<- naiveBayes(Personal.Loan ~ CreditCard + Online, data = training_data)

new_customer<- data.frame(CreditCard = 1, Online = 1)

predict(nb_model, new_customer, type = "raw")
```

```
##           0           1
## [1,] 0.8843065 0.1156935
```

Running Naive Bayes on the data gave me 0.1157 or 11.57% for $P(\text{Loan} = 1 \mid \text{CC} = 1, \text{Online} = 1)$, while the probability I obtained in Part E was 0.1106 or 11.06%. The two values are very close and only differ by 0.52 percentage points. This indicates that manual calculation and the Naive Bayes model produce consistent results.